Bello. B

FEDERAL UNIVERSITY LAFIA
DEPARTMENT OF PHYSICS
FACULTY OF SCIENCE
FIRST SEMESTER EXAMINATION 2017/2018 SESSION

PHY111 (General Physics I) Credit Unit: 3

Instruction: Answer ALL questions.
Clearly circle the correct option with your pen/pencil.

TIME ALLOWED: 2.5 HRS

## SECTION A

1). One of these is not a fundamental force
(A) Strong Force (B) Gravitational Force (C) Normal Force (D) Weak Force

2). The unit for angular velocity is
(A) $Nm^{-1}$ (B) $Nms^{-2}$ (C) $ms^{-2}$ (D) $rads^{-1}$

3). A photographs records at $33\frac{1}{3}$ revolution per minute. Express the angular velocity
(A) $34.9Nm^{-1}$ (B) $34.9Nm^{-2}$ (C) $3.49\ rads^{-1}$ (D) $0.349ms^{-1}$

4). Circular motion is that which causes day and night on earth whereas rotational motion causes ...................................... around the year.
(A) Season (B) Motion (C) Cloud (D) Weather

5). According to Hook's law $F = -kx$, $k$ is known as
(A) Elastic Constant (B) Inelastic Constant (C) Bulk Modules Constant
(D) Limit exceeded Constant

6). There are broadly two types of deformation one of which is
(A) Plastic deformation (B) Central deformation (C) Local deformation
(D) Continuous deformation

7). A force of 0.8N stretches an elastic spring by 2cm. find the spring constant of the spring.
(A) $20Nm^{-1}$ (B) $40Nm^{-1}$ (C) 4.0kg (D) $0.2kgN^{-1}$

8). A particle is moving in a circle of radius r, in time t, with a uniform speed v. The angular displacement is
(A) $V^2/r$ (B) Vr (C) r/V (D) V/r

9). Inelastic materials are materials that
(A) Obey hooke's law (B) Within elastic limit (C) exceeded elastic limit
(D) Gravitational force acted

10). The magnitude of centripetal acceleration is given by
(A) $V^2/r$ (B) $r^2/V^2$ (C) $V/r^2$ (D) V/r

11). A robot is initially at 2.20m/s along a hallway in a space terminal. Its speed increases to 4.8m/s in time 0.20sec. Determine the magnitude of it's average acceleration along the path traveled.
(A) $13m/s^2$ (B) $15m/s^2$ (C) $12m/s^2$ (D) $11m/s^2$



$\frac{V_1 - V_2}{t}$

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2). Newton's law of universal gravitation $F = G\frac{M_1M_2}{R^2}$. The value of $G$ is given as
(A) $1.67 X 10^{-11} kg^{-1}m^3s^2$ (B) $9.8ms^{-2}$ (C) $6.67 X 10^{-11} kg^{-1}m^3s^{-2}$ (D) $8.85 X 10^{-12}$ [illegible]

3). Young modulus is also known as
(A) Elastic Modulus (B) Stress Modulus (C) Plastic Modulus (D) Coefficient Modulus

4). A 75kg weight stands on a compression spring with a spring constant 5000N/m and normal length 0.25m, what is the total length of the loaded spring ($g = 10 m/s^2$).
(A) 0.15m (B) 0.10m (C) 0.40m (D) 0.25m

5). Calculate the gravitational force of attraction of two cars 5m apart if the masses of the cars are 1200kg and 1200kg ($G = 6.7x10^{-11} Nm^2kg^{-2}$).
(A) $13.2x10^{-6}N$ (B) $2.3x10^{-6}N$ (C) $32x10^{-6}N$ (D) $30x10^{-6}N$

## SECTION B

1). A 500kg car is driving in a 25m circle at $18ms^{-1}$ calculate the centripetal force.
(A) 6500N (B) 6350N (C) 6040N (D) 6300N

2). Motion of very small atomic and sub-atomic particles is described in
(A) Classical mechanics (B) Relatives mechanics (C) Quantum mechanics
(D) Quantitative mechanics

3). Assuming that we only know the period of oscillation of a simple pendulum, T which depends on its mass, m, the length, L and the acceleration due to gravity, g. determine the exact form of the equation using dimensional analysis
(A) $T = k\frac{L^{\frac{1}{2}}}{g}$ (B) $T = k\left(\frac{L}{g}\right)^{\frac{1}{2}}$ (C) $T = k\frac{L^2}{g}$ (D) $T = k\frac{L}{g^2}$

4). Suppose that the displacement, S of an object is related to time, t according to the expression $S = Bt^2$. What is the dimension of B
(A) $V^2L^{-1}$ (B) $LT^{-1}$ (C) $LT^{-2}$ (D) $LT^2$

5). The following equation was given by a student during an examination $v = v_0 + at^2$. Where $v$ and $v_0$ are the final and initial velocity, $a$ is the acceleration and t is time. Is the equation correct?
(A) No (B) Yes (C) Not Sure (D) All of the above

6). Given two vectors $\vec{F}_1$ and $\vec{F}_2$. $\vec{F}_1$ is 6 units long and makes an angle of $36^0$ with the x-axis, $\vec{F}_2$ is 7 units long and is in the direction of the negative x-axis. Find the vector sum, R of the two vectors.
(A) 74.2 units (B) 8.61 units (C) 4.13 units (D) 12.0 units

7). What is the angle $\theta$ between vectors $\vec{A} = 3i - 4j$ and $\vec{B} = -2i + 3k$
(A) $cos^{-1}\left(\frac{-6}{5\sqrt{13}}\right)$ (B) $sin^{-1}\left(\frac{-6}{5\sqrt{13}}\right)$ (C) $sin^{-1}\left(\frac{-6}{5\sqrt{13}}\right)$ (D) $tan^{-1}\left(\frac{-6}{5\sqrt{13}}\right)$

8). Given two vectors $\vec{A} = 4i - 2j - 3k$ and $\vec{B} = 2i - j + 6k$. Calculate vector $\vec{D}$ such that $2\vec{A} - 3\vec{B} + \vec{D} = 0$.
(A) $\vec{D} = -2i + j + 24k$ (B) $\vec{D} = 2i - j + 24k$ (C) $\vec{D} = -2i + j + 12k$
(D) $\vec{D} = 2i - j - 24k$

27). The coefficient of volume expansion of water between 0 $^oC$ and 4 $^oC$ is ..........

(A) Unchanged (B) positive (C) zero (D) negative

28). If 8950 $J$ of heat is added to 3.00 $moles$ of iron; What is the temperature increase of the iron? (molar heat capacity of iron = 26.3 $J/mol.K$).

(A) 113.4 $^oC$ (B) 340.3 $^oC$ (C) 8.8 $^oC$ (D) 78.9 $^oC$

29). How much heat is required to convert 12.0 $g$ of ice at -10.0 $^oC$ to steam at 100 $^oC$? (specific heat capacity of water = 4190 $J/kg.K$; Latent heat of vaporization = 2256x10$^3$ $J/kg$; Latent heat of fussion = 334x10$^3$ $J/kg$).

(A) 27072.8 $J$ (B) 36610.8 $J$ (C) 27574.8 $J$ (D) 5530.8 $J$

30). A 4.00 $kg$ iron is taken from a furnace, where its temperature is 750 $^oC$ and placed on a large block of ice at 0 $^oC$. Assuming that all the heat given up by the iron is used to melt the ice, how much ice is melted? [Latent heat of fussion of ice = 334x10$^3$ $J/kg$; specific heat capacity of iron = 440 $J/kg.K$]

(A) 3.95 $kg$ (B) 2.10 $kg$ (C) 1.41 $kg$ (D) 4.22 $kg$

31). Thermal energy is moved from the lower atmosphere to the upper atmosphere through the process of

(A) radiation (B) convection (C) conduction (D) transfer

32). A 0.02 $m^3$ tank contains 0.225 $kg$ of helium at 18.0 $^oC$. If the molar mass of helium is 4.00 $g/mol$, what is the pressure in the tank? (specific gas constant = 8.314 $J/mol.K$)

(A) 86.0x10$^5$ $Pa$ (B) 68.0 x10$^5$ $Pa$ (C) 4.2 x10$^5$ $Pa$ (D) 1088.7 x10$^5$ $Pa$

33). A room becomes warmer by 10 K. How much warmer has it become in degrees Fahrenheit?

(A) 18 $^oF$ (B) 5.6 $^oF$ (C) 10 $^oF$ (D) 273.15 $^oF$

34). Thermal propagation through solids is by:

(A) conduction (B) convection (C) radiation (D) transfer

35). Boyle's law states that

(A) $Pressure \propto Volume$ (B) $Pressure \propto \frac{1}{Volume}$

(C) $Volume \propto Temprature$ (D) $Volume \propto \frac{1}{Temprature}$

9). Given that $\vec{A} = -i + 2j - 2k$. Find the unit vector in the direction of $\vec{A}$

(A) $-i/_3 - 2j/_3 - k$ (B) $-i/_3 + 2j/_3 - 2k/_3$

(C) $i/_3$ [illegible] $/_3 + k$ (D) $-i/_3 + 2j + k$

10). Suppose the motion of a particle is described by the equation $x = a + bt^2$ ; where $a$ = 20m, and $b$ = 4$ms^{-2}$. Find the average velocity of the particle in time interval between $t_1$ = 2s and $t_2$ = 5s.

(A) 84.0 $ms^{-1}$ (B) 2.0 $ms^{-1}$ (C) 28.0m (D) 28.0$ms^{-1}$

11). The distance travelled by a particle starting from rest is plotted against the square of the time elapsed from the commencement of the motion. The resulting graph is linear. The slope of the graph is a measure

(A) Initial displacement (B) initial velocity (C) acceleration (D) half acceleration

12). A stone is released from rest from the top of an inclined platform, requiring 5s to cover a distance of 100cm in falling down to the ground. What distance will it have falling vertically in the same time.

(A) [illegible] (B) 122cm (C) 122mm (D) 50m

13). A constant force of 5N acts for 5 seconds on a mass of 5kg initially at rest. Calculate the final momentum.

(A) 25$kgms^{-1}$ (B) 0$kgms^{-1}$ (C) 125$kgms^{-1}$ (D) 5$kgms^{-1}$

14). When taking a penalty kick, a footballer applies a force of 30.0N for a period of 0.05s. If the mass of the ball is 0.075kg, calculate the speed with which the ball moves off.

(A) 4.50$ms^{-1}$ (B) 11.25$ms^{-1}$ (C) 20.00$ms^{-1}$ (D) 45.00$ms^{-1}$

15). A body of mass 100g moving with a velocity of 10.0$ms^{-1}$ collides with a wall. If after the collision, it moves with a velocity of 2.0$ms^{-1}$ in the opposite direction, calculate the change in momentum.

(A) 0.8Ns (B) 1.2Ns (C)12Ns (D)80.0Ns

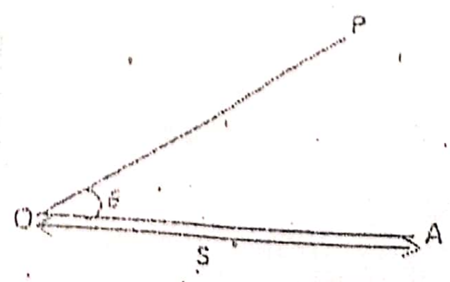
16). A man of mass 50kg ascends a flight of stairs 5m high in 5s, if the acceleration due to gravity is 10$ms^{-2}$. calculate the power expended

(A) [illegible] (B) 200w (C) 400w (D) 500w

17). Under which of the following condition is work done

(A) a man supports a heavy load above his head with his hand (B) a man holds olds a child on his hand for 2 hours (C) a boy climbs unto a table (D) a woman holds a pot of water for 10 hours

18). A boy pulls an object O with a force P at distance S along a line OA acting at an angle $\theta$ as shown What is the work done (W) by the boy?



(A) $W = P \times S$ (B) $W = P \sin\theta \times S$ (C) $W = P\cos\theta \times OP$

(D) $W = P\cos\theta \times S$

19). A boy of mass 10kg climbs up 10 steps each of height 0.2m in 20s. Calculate the power of the boy

(A) 10W (B) 200W (C) 1000W (D) 20W

20). The maximum velocity of a swinging pendulum bob of mass m when its kinetic energy is maximum is:

(A) $v_{max} = \sqrt{2gh}$ (B) $v_{max} = \sqrt{gh}$ (C) $v_{max} = \frac{\sqrt{gh}}{2}$

(D) $v_{max} = 2gh$

21). Two systems are in thermal equilibrium if and only if ....................?

(A) they have energy (B) they have the same temperature

(C) they are in thermal contact (D) they are heated

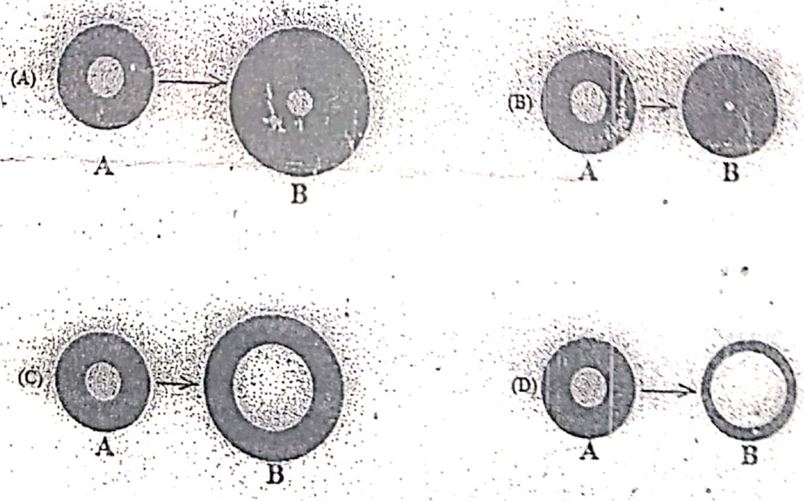
22). The temperature at which all molecular/atomic motion stops is

(A) 100 °C (B) 0 °C (C) 273 °C (D) -273 °C

23). If the air temperature is given as 115 °F, what will this temperature be on centigrade scale?

(A) 46 °C (B) 15 °C (C) 32 °C (D) 388 °C

24). If an object in A is heated and it expands to B, which of the diagrams below will be correct?




25). A material has a coefficient of volume expansion of $3.6 \times 10^{-5}\ K^{-1}$. What is the coefficient of linear expansion of the material?

(A) $1.2 \times 10^{-4}\ K^{-1}$ (B) $1.08 \times 10^{-4}\ K^{-1}$ (C) $1.2 \times 10^{-5}\ K^{-1}$ (D) $1.08 \times 10^{-5}\ K^{-1}$

26). A glass flask with volume 200 $cm^3$ is filled to the brim with glycerin at 25 °C. How much glycerin over-flows when the temperature of the system is raised to 105 °C? (linear expansivity of glass = $0.40 \times 10^{-5}\ K^{-1}$; cubic expansivity of glycerin = $49 \times 10^{-5}\ K^{-1}$)

(A) 2.700 $cm^3$ (B) 7.648 $cm^3$ (C) 7.840 $cm^3$ (D) 0.192 $cm^3$